

ABHISHEK CHATTERJEE

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Summary AI/ML Engineer and BTech CSE (AI & ML) fresher with internship and project experience in Python, Machine Learning, Deep Learning focused on building scalable AI systems. Interested in solving real-world industrial problems and delivering impactful, data-driven solutions. Skills

- Programming Languages & Frameworks: Java | C++ | C | Python | HTML | CSS | JavaScript | Node.js | Express.js | React.js | TypeScript.js
- Database, Tech skills & Tools: MySQL | MongoDB | Generative AI | Machine Learning | Git | VS Code | NumPy | Pandas | Scikit-learn | TensorFlow | Google Collab | RESTful APIs | FastAPI | Deep Learning | Computer Vision | PyTorch | OpenCV | Kornia | Remote Sensing | Google Cloud Platform

Education

- **Sarala Birla University, Ranchi, Jharkhand** **2022- 2026**

(BTech) Bachelor of Technology in Computer Science (Specialization: AI&ML) | CGPA: 8.33/10

- Mother's International School, Ranchi, Jharkhand (Senior Secondary (Class XII) – CBSE) 2022
- Shanti Rani School, Ranchi, Jharkhand (Secondary (Class X) – JAC) 2020

Work Experience

ISRO (NRSC), Shadnagar | Software Intern
2026- Jun 2026

Mar

- Trained and fine-tuned custom DeDoDe (Detector-Descriptor) models using PyTorch and Kornia on NISAR SAR HDF5 remote sensing datasets.
- Enhanced feature detection, descriptor learning, and image matching performance by 20%+, improving keypoint correspondence accuracy for geospatial analysis.
- Developed an AI-driven PDF-to-LaTeX converter that automated scientific document reconstruction, reducing manual effort by 85%+.

- Optimized SAP Material Management master data and reporting processes, increasing workflow efficiency and data retrieval speed by 20% while improving operational decision-making by 15%.

Projects Plant Disease Recognition System | Github_Link Nov 2025

[Python, TensorFlow/Keras, Flask, NumPy, scikit-learn, Pandas, Matplotlib, Seaborn, Jupyter Notebook]

- Developed a deep learning CNN-based computer vision model using TensorFlow/Keras to classify 38 plant diseases with 95% validation accuracy on 87,000+ images across multiple crops.
- Built an intelligent crop health monitoring system for early disease detection, helping farmers reduce crop damage by 20%.

PDF-to-LaTeX Reconstruction Engine | Github_Link May - June 2026

[FastAPI, React, CodeMirror 6, PyMuPDF, OpenCV, KaTeX, WebSockets, Docker]

- Built an AI-powered PDF-to-LaTeX conversion platform using FastAPI, React, PyMuPDF, and OpenCV, achieving 90%+ layout fidelity in complex document reconstruction.
- Created a custom LaTeX editor with real-time rendering, syntax diagnostics, and live preview capabilities, reducing manual verification effort by 70%+.

Deepfake Detection System | Github_Link

Sept 2025

[Python, OpenCV, NumPy, Scikit-learn, Streamlit, Matplotlib, Seaborn, tqdm]

- Designed an interactive deepfake authentication application achieving 90%+ accuracy in detecting manipulated media through advanced computer vision techniques and statistical analysis pipelines.

[Python, Scikit-learn, Pandas, NumPy, Streamlit]

- Achieved 97% ROC-AUC by training a Random Forest on an imbalanced dataset with SMOTE and preprocessing.
- Deployed a Streamlit app for real-time fraud prediction, boosting detection accuracy by 45% over baseline.

Coursework

- Object Oriented Programming System
- Software Engineering
- Deep Learning
- Introduction to Artificial Intelligence
- Bio-Inspired Computing
- Database Management System
- Machine Learning
- Natural Language Processing
- Data Structures and Algorithms
- Theory of Computation
- Knowledge Engineering & Expert Systems
- Operating System
- Computational Intelligence
- Computer Networks
- Advanced Artificial Intelligence

Training, Certifications, Achievements & Roles

- Open Source Head, Google Developer Groups (GDG) On Campus, Sarala Birla University.
- Data structure and Algorithms in Java Certification from Algoprep.
- CCNA: Introduction to Networks, Python Essentials 1& 2 Certification from CISCO Networking Academy.
- Smart India Hackathon Finalist 2024 (Top 2%) – Developed a game "AquaGuardian" focused on water conservation.
- Campus Ambassador, Contributor at GirlScript Summer of code 2025(Gssoc).
- IBM SkillsBuild PBL 2025 – AI Agent Architect Certification
- Infosys Springboard Certifications (Introduction to DS, Computer Vision, Introduction to DL, Introduction to AI & more).